

Starry Starry Night: An Interactive Narrative Visualization of Astrology Imagery in Tang Poetry

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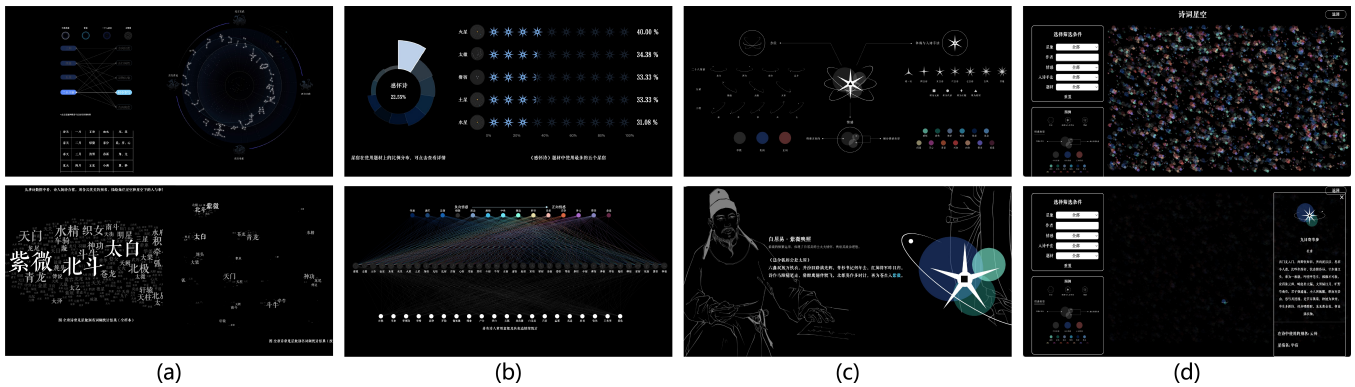


Figure 1: An overview of *Starry Starry Night*, including representative visualizations of (a) Chapter One, (b) Chapter Two, (c) Chapter Three, and (d) Exploration Page.

ABSTRACT

Astrology emerges as a distinctive and mystical facet of traditional Chinese culture, imparting a unique charm when interwoven into ancient literature such as Tang poetry. We designed an interactive narrative visualization, *Starry Starry Night*, to present the application of astrology imagery in Tang poems. The visualization combines author-driven storytelling and reader-driven exploration, hoping to effectively enhance readers' understanding of astrology-related poetry and to promote the dissemination of traditional Chinese culture.

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1 INTRODUCTION

Narrative visualizations communicate data insights by telling a story [Segel and Heer 2010]. They are effective for knowledge dissemination, enabling authors to share insights with their audience [Dove and Jones 2012]. Narrative visualizations have been used in disseminating scientific knowledge, publishing political news, and promoting traditional culture. Current narrative visualizations focusing on traditional culture [Li et al. 2021][Chen et al. 2018] mostly use linear narrative approaches. These approaches can be categorized as author-driven and reader-driven [Segel and Heer 2010]. Author-driven visualizations follow a strict linear path with abundant messaging, while reader-driven visualizations have no prescribed order and offer high interactivity. Blending both approaches caters to predetermined storytelling by the author and personalized discovery by the reader. Our work highlights the importance and application of astrology in Tang poetry through interactive narrative visualization. We combine author-driven storytelling with reader-driven data exploration, enabling readers to comprehend the astrological elements in Tang poetry while exploring specific poems based on individual interests. This approach enhances the dissemination of traditional Chinese culture. By integrating these approaches, we provide a richer, more engaging experience, allowing users to delve deeper into the intricate patterns and emotional nuances of Tang poetry. It can be accessed at <https://astrologytangpoetry.github.io/>.

Dimensions					
Constellations Related	Constellation Names		Constellation Alias	Position And Area	Symbolic Meaning
Poetry Related	Title	Author	Content of verse	Format	Genre
				Five-character regulated verse, seven-character quatrain, ancient style, lyrics...	Landscape and pastoral poetry, philosophical poetry, farewell poems, homesickness poems during a journey...
The Application Of Constellations In Poetry	Technique			Emotions (Positive or negative)	Subcategory of emotions
	· Stars as the main theme · Stars as literary elements · Stars as symbolic representations · Stars as references or metaphors				Excitement, Love, Sadness, Confusion, Anger, Longing, Admiration, Enlightenment...

Figure 2: The dimensions of poem data.

2 OUR APPROACH

2.1 Data Processing

The data processing involves three steps: *data collection*, *data filtering*, and *data analysis*, resulting in a dataset of 2,385 Tang poems for visualization.

Data Collection. We used an open-source poetry database [Xiuze 2022], focusing on its Tang poetry section. Each poem includes four dimensions: *Title*, *Author*, *Format*, and *Content of verse*.

Data Filtering. First, we gathered aliases of each constellation from literature and identified 3,430 poems containing constellation aliases using a keyword filtering script. We manually verified these references, ultimately filtering out 2,385 poems associated with astrology.

Data Analysis. Based on research on astrology in poetry, we analyzed each poem, adding dimensions like *Genre* and *Techniques* of incorporating astrological aliases. Given the importance of emotion in poetry, we focused on analyzing emotions using NLP analysis [Baidu 2023] to determine positive or negative sentiments. Manual annotation further categorized emotions into fourteen subcategories, such as excitement, anger, and serenity. This resulted in a Tang poetry dataset with 2,385 poems, each associated with 12 distinct dimensions, as shown in Figure 2.

2.2 Visualization Design

Based on the above dataset, we have developed an interactive narrative visualization, *Starry Starry Night*, comprised of an author-driven storytelling section and a reader-driven data portrait exploration section, as shown in Figure 1.

2.2.1 Author-Driven Data Storytelling. We conducted a cross-analysis of the dataset and developed a data story in three chapters. Chapter One provides an overview of ancient astrological theories using a diagram. Word clouds depict the frequency of constellations and their aliases in Tang poetry. Chapter Two explores how constellations influence Tang poetry. Donut, bar, and rose charts illustrate the distribution of themes, techniques, and emotions in poems related to different constellations. A node-link graph explains Tang dynasty poets' preferences in using constellations and expressing emotions. Chapter Three showcases data portraits of representative poems and poets. We visually encoded the dimensions of each poem and generated data portraits for all 2,385 poems, selecting four representative poets and one notable poem each. The specific data portraits are displayed, describing the relationship between the poem, the preferred constellation, and the poet.

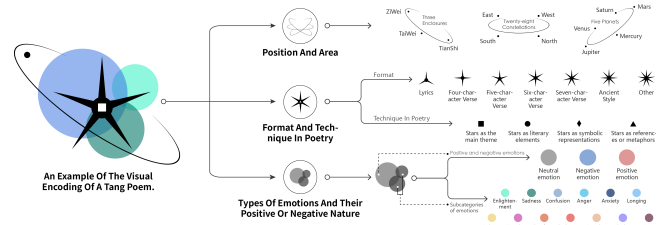


Figure 3: The visual encoding of a poem “star”.

2.2.2 Reader-Driven Data Exploration. We designed an interactive “starry sky of poems”, allowing readers to explore the data portraits of all poems. The 2,385 poems are presented as twinkling stars. Users can click each star to view detailed information and dimensions of the corresponding poem. A panel on the left side provides filtering options across various dimensions, enabling users to precisely explore poems that match their interests. In terms of visual encoding, we designed data portraits for the stars, adopting the image of a star. Different encoding strategies are applied based on data attributes, as shown in Figure 3. Visual attributes related to quantity represent numerical dimensions; for example, the number of angles on a star corresponds to the character count of a verse. Different shapes encode dimensions categorized by type, and color attributes represent discrete or continuous values, conveying emotional tones. Warm and cool colors indicate positive and negative emotions, with specific colors for each subcategory of emotions.

3 SUMMARY AND FUTURE WORK

In summary, we have developed an interactive narrative visualization that integrates author-driven storytelling and reader-driven exploration to illustrate the application of astrology in Tang poetry. Our work demonstrates how visualizations can effectively convey and promote the understanding of traditional culture. In our work, we have currently combined the two visualization approaches in a relatively simple manner. In the future, there is potential for extending the spectrum of the forms of combining author and reader-driven visualization. For instance, data portraits could emerge as components of narrative storytelling. Also, how to optimize the visual encoding of data portraits to make information comprehension more easily and efficiently, is worth discussing too.

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